

Topic: Visualising spatial data

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Unit: FIT5147 Data exploration and visualisation - S2 2026

Book: Topic: Visualising spatial data

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1. Introduction

In this topic, you will learn about the design and implementation of data maps: from the available geospatial data formats used in common software related to Geographical Information Systems (GIS), to the actual design of maps in terms of coordinate systems, projections and common uses of marks and channels to derive map idioms. You will learn about the common types of maps and finally how to provide data analytics via the use of maps for geospatial data, with general guidelines on how to communicate common statistics such as aggregation, clustering, correlation, uncertainty, etc.



Exploring the learning materials

As you work through the learning materials in this Moodle book, please complete the learning activities and consider the reflection opportunities presented. Various videos and webpages may be included for your reference to reinforce and expand on the contents of this book. You are recommended to watch any videos produced by Monash University. Unless stated otherwise, the other resources are optional but you may find them very helpful. If any links no longer work, try searching for the title elsewhere.



Topic readings

The following resources are good introductions to the design of effective data maps. They are optional reading that would supplement your understanding of this book.

- [Thematic cartography and geovisualization](#) , 4th edition (Slocum, McMaster, Kessler and Howard, 2022) - accessible [via the Monash University library](#)
- [Cartography: Thematic Map Design](#) (Dent, Torguson and Hodler, 2009).

2. Visualising spatial data

Watch the following video which discusses the background for visualising spatial data.

Select the advanced player features ^ control to access the video table of contents with chapters targeting each topic.

NOTE: This video is a little old. The comment about "two weeks" is no longer correct as we have combined the two weeks into 1 week for the FIT5147 taught material.

FIT5147_M3.1 Visualising Spatial Data



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Visualising spatial data (2022) Monash University <[Panopto Viewer](#)>

Resources

You might be looking for these resources referenced in the video.

John Snow, cholera map 1854

- [John Snow's data journalism: the cholera map that changed the world, The Guardian](#) (Rogers 2013).
- [John Snow's cholera map of London recreated Guardian Data Blog](#) (Rogers 2014).

Geographical Information Systems (GIS)

- [GIS \(Geographic Information System\)](#) (National Geographic 2013).

3. Spatial data formats and tools

Now a wide range of tools can be used to create maps. Business intelligence tools like Tableau provide basic mapping functionality, while R and Python both have packages for creating maps. Those in R include **ggmap** and **Leaflet**. However, if you need to do a lot of spatial data analysis, you should consider using specialised Geographic Information Systems (GIS) software.

However, the first step is to procure data that is geospatial in nature, data that references locations within a particular coordinate system (e.g. latitude and longitude points).

Watch the following video, which discusses the format and tools for spatial data, including GIS tools and resources and GIS data formats.

Select the advanced player features ^ control to access the video table with chapters targeting each topic.

FIT5147_M3.6 Format and Tools for Spatial Data



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Format and tools for spatial data (2022) Monash University <[Panopto Viewer](#)>

3.1. Geographic Information Systems

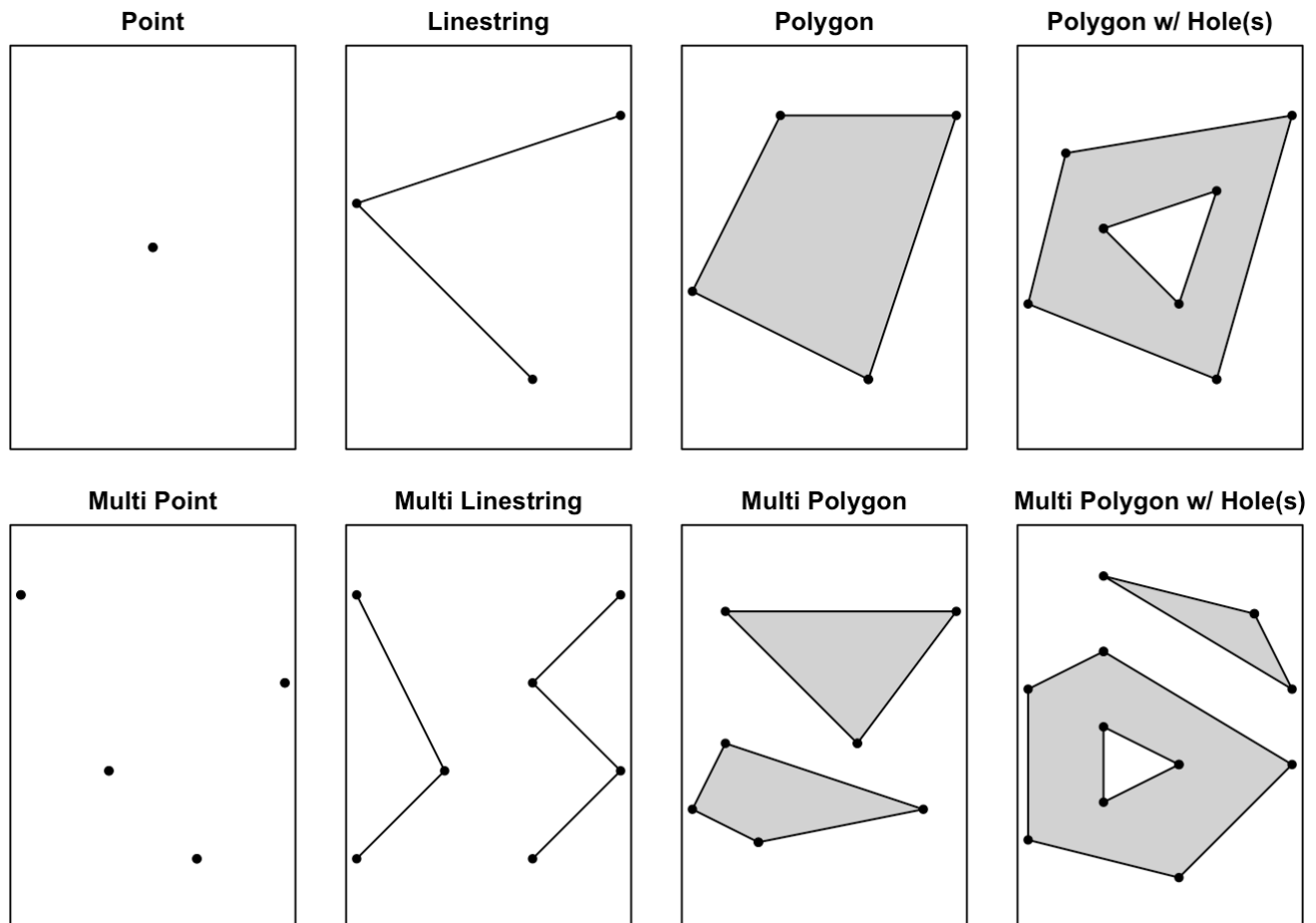
Geographic Information Systems (GIS) date from the late 1960s. A modern GIS can be used to capture, store, transform, analyse and visualise spatial or geographical data. Data maps are created by creating different layers for different data objects, such as towns, borders or rainfall. Elements in the layers have spatial and sometimes temporal coordinates, and the layers are overlaid to form the data map. GIS typically provide good spatial analysis tools and allows a smooth transition between different levels of detail. Some GIS software solutions are:

- [Esri ArcGIS](#) (Esri n.d.) software is the industry standard in commercial GIS systems. It allows users to combine a wide variety of data sources to create data maps, comes with cartographic data and can perform spatial analysis.
- [Mapbox](#) (Mapbox n.d.), [CartoDB](#) (Carto 2023), [MapInfo](#) (Precisely 2023) are other commercial GIS tools. The first two currently offer limited functionality for free.
- [QGIS](#) (QGIS 2022) is an open-source GIS tool.
- Google provides Google Fusion Tables, Google Charts and APIs to Google Maps and Google Earth.
- [National Map](#) (Australian Government 2021) was a great online GIS tool for quickly viewing Australian government data. It was replaced in 2025 by the [Digital Atlas of Australia](#).
- [Open Street Map \(OSM\)](#) (Open Street Map 2018) is a great online open-source resource for maps.

3.2. Spatial data formats

Spatial data is stored in a GIS system (or found as standalone files) in two ways:

- As **raster** or pixel-based data. Aerial photos, satellite images, and elevation data are stored as raster data.
- Geographic **shapes** are typically stored as vector data. **Points** are used to store the location of zero-dimensional features like a mountain peak, **lines** or poly-lines to store the position of one-dimensional features like roads or rivers, and **polygons** to store the position and shape of two-dimensional features like lakes or countries.



Types of shape files based on geometry. This applies to most shape file formats (non-raster) like GeoJSON. [Source](#).

A wide variety of different formats are used by GIS software to exchange spatial data. [JPEG2000](#) (Wikipedia 2022) or [GeoTIFF](#) (Wikipedia 2022) are often used for GIS raster data. Four common formats for GIS vector data exchange are:

- [Keyhole Markup Language \(KML\)](#) (Wikipedia 2022) – XML-based open standard that underlies Google Maps.
- [Geography Markup Language \(GML\)](#) (Wikipedia 2022) – XML-based open standard developed by the Open Geospatial Consortium that includes raster data.
- [Esri shapefile](#) (Wikipedia 2023) – widely used format developed by Esri.
- [GeoJSON](#) (Wikipedia 2023) – JSON format used by many open-source GIS packages.

Some of these formats are more accessible than others. For example, certain formats like GML or KML require specialised software to open and edit them. When it comes to public repositories of spatial data, formats like ESRI or GeoJSON are more common since they can be opened and edit by multiple software suites, including popular ones like Tableau, R and D3.js.

There are tools to convert between these different formats. Vector data may also be specified using [SVG](#), the W3C Scalable Vector Graphics standard. One such tool is [mapshaper.org](#), excellent, easy to use, browser-based application that supports multiple formats like ESRI, GeoJSON, TopoJSON, SVG. It allows users to visualise geospatial data, filter, edit and convert it to other formats.

4. Map design fundamentals

Here we will look at the various decisions and steps involved in designing a data map (or thematic map as cartographers usually call them). The first question is whether you should actually be using a map. In short, you should use a map if the data is associated with a geographic location or region and you wish to explore or show how this affects the other attributes. Just because data has a geographic location doesn't necessarily mean you should show it on a map. For instance, if you are only interested in comparing sales figures for different stores in a company, you would better compare these using a bar chart. However, if you are interested in seeing how the location of the store affects sales figures, then you should use a map. This might, for instance, show sales figures for the stores overlaid on a choropleth map of average income broken down by suburb.

Maps are one of the more difficult visualisations to design: inadvertently creating a misleading map is easy.

4.1. Scale and generalisation

Maps can be drawn at very different levels of scale. The scale of a map is often expressed as a ratio of map units to Earth units. This is called the *representative* fraction. For instance, a scale of 1:10,000 means that 1 cm on the map represents 10,000 cm = 100 m on the earth's surface. Map scale can be shown using a representative fraction or a graphical bar scale or even given verbally, as in '1 inch to the mile.' Graphical bar scales are by far the most easily understood way of showing the scale of a map and should be included whenever the scale is relevant to understanding the data, and the scale is relatively consistent over the map.

The scale of a map dictates how much detail can be shown. A *large scale* map shows a small area in detail, while a *small scale* map shows a large area in less detail. Spatial data sets often contain very detailed information. One part of map design is working out the appropriate level of detail to show in the map and then aggregating or generalising the geographic and non-geographic data appropriately. For instance, imagine you wish to compare population density. On a map of the world, you might show country boundaries, on a map of Australia you might show state boundaries and the capital cities.

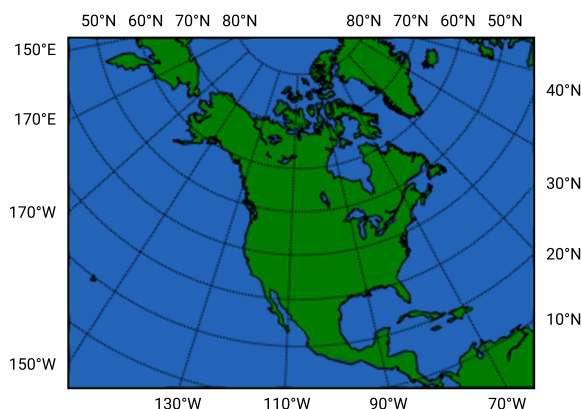
4.2. Coordinate system

The Earth roughly speaks like a sphere though it bulges a bit at the equator because of the centrifugal forces generated by rotation. Location on the earth's surface is given by latitude and longitude. The location of Melbourne is given by 37°48'49"S (latitude) and 144°57'47"E (longitude).



Spherical Earth (2022) Monash University

Latitude lines go from east to west and are parallel to the equator. They run from 90°N which is the North Pole to 0° which is the equator down to 90°S which is the South Pole. Longitude lines run at right angles to the latitude lines and are circles running north to south through the poles. Longitude is measured east or west from Greenwich in the UK. They range from 180°W to 0 which is Greenwich to 180°E (which is, of course, equal to 180°W). Probably like many, you can't remember which is which: lines of longitude are all "long" while lines of latitude vary in size. Lines of constant latitude are called parallels as they are parallel to the equator. Lines of constant longitude are called meridians, and the meridian through Greenwich is called the Prime Meridian.



Longitude and latitude (2022) Monash University

4.3. Map projections

One of the great difficulties facing map-makers is how to faithfully show the surface of the Earth (a flattened sphere) on a flat piece of paper. A *map projection* is the mapping between points on the surface of the Earth and the position on the two-dimensional map. You might like your map projection to avoid distorting angles, distances, areas and directions. Unfortunately, this is impossible: there is no map projection that can do this.

The distortions are small for maps showing only a tiny part of the Earth's surface, but the distortions are considerable for maps showing continents or the entire world. As a result, hundreds of different map projections have been invented. There is no single best projection: the appropriate choice depends upon the task, the area covered by the map and the kind of map. Thus, choosing the right projection is important in designing a map.

Projections can be classified depending on which property they preserve

- **Equal area (or equivalent)** projections preserve the relative size of areas.
- **Conformal projections** preserve angles locally.
- **Equidistant projections** preserve distance from a particular location or along a particular direction.
- **Compromise projections** neither preserve area nor angles, but ensure that area and angle distortion is 'not too bad' overall.

The most commonly used map projection is that of Mercator, because Google Maps, Apple Maps and many other online maps use it. Gerard Mercator invented this projection in 1569. His map was designed to help European sailors who urgently needed nautical charts that allowed them to navigate between very distant locations easily. His map uses a grid of perpendicular meridians and parallels. Mercator's projection ensured that a straight line drawn between two points on the map gives the correct bearing in which to travel between the points. This line is not the shortest connection between the two points, but this characteristic of the Mercator projection is extremely useful for marine navigation. However, it comes at a price: the scale is not consistent across the map, and regions at high latitudes are hugely inflated. For example, the area of Greenland is 14 times enlarged, making it appear the same size as Africa. Furthermore, it is not even possible to show the North or South Pole. Because of the serious, erroneous conceptions the Mercator projection creates, it should not be used for world maps or maps showing areas of the size of continents. Nonetheless, the Mercator projection has become the standard map projection used on many maps, including those of Google.

Table 3.2 from *Cartography: Thematic Map Design* provides a guide to choose a projection for different purposes and regions. Alternatively, use [Projection Wizard](#). Because map projections cannot preserve all properties (such as areas, angles, distances) of the original sphere, map authors must consider which properties are the most important to retain. This consideration is based on the purpose of the map and the map type used to visualize information. Aesthetic appearance of the map, aspect ratio, and whether available software supports a projection are additional criteria for selecting a projection. Many of the recommended projections below minimize area distortion, which is often a useful property to preserve.

A reasonable choice for maps of the *world* is the **Robinson** projection, as most readers find it pleasing-looking. An alternative is the **Winkel Tripel** projection. The Robinson and the Winkel Triple projection are both compromise projections that reduce both area and angular distortion. However, if you are creating a map of the world in which area is important in interpreting the data, such as when using a dot map to show total population and density, then an equal-area projection should be used. Good equal-area projections for world maps are the **Equal Earth** projection, which looks similar to the Robinson projection, and the elliptical **Mollweide** projection.

For showing *continent-sized areas and smaller areas*, a good choice is the **Lambert Azimuthal Equal-area** projection appropriately centred on the region, especially if the format of the mapped area is close to a square. If the mapped area has landscape format, then the **Albers Equal-area Conic** projection is a good choice when mapping an area at a medium latitude. If the mapped area is elongated and follows a great circle on the sphere (such as the equator or any meridian), then a **Cylindrical Equal-area** projection is an appropriate choice.

Azimuthal, conic and cylindrical are three categories commonly used in cartography to classify projections for areas smaller than the entire world. The conceptual idea is that the sphere is first projected onto a developable surface, and then this surface is "developed" into a flat area. The three existing developable surfaces are cylinder (→ cylindrical map projection), cone (→ conic map projections) and a plane (→ azimuthal or planar map projections).

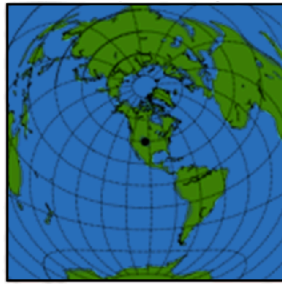
Azimuthal equidistant projection**Mollweide projection****Mercator projection****Robinson projection****Lambeth azimuthal equal area projection**

Table 3.2 from Cartography: Thematic Map Design (2022) Monash University

It is worth emphasising that when choosing the map projection, you can also choose where to centre the map (or developable surface) and the point or directions for which distortion is minimised: obviously, you should choose these to minimise distortion of the most important areas. For smaller regions, the choice of map projection is not as important as the amount of distortion is less (hence the Mercator projection is an appropriate choice for large-scale maps showing small areas).

Given the problems with projection, it is sometimes worth thinking about another approach: actually showing the Earth as a three-dimensional globe. The disadvantage of this is the difficulty of navigating in 3D space and that curvature means it is impossible to see more than half the globe at once, and that depth introduces difficulties in accurately comparing distances and areas. However, it does have the advantage that there is bias added by a map projection.

4.4. Map Projections 1

Purpose

This activity is intended to check your understanding of different map projections.

Complete the following questions by selecting the correct response from the dropdown list.

Match the projection type with the correct property that is being preserved.

-  **projections** (or equivalent) preserve area.
-  **projections** preserve angles locally.
-  **projections** projections preserve distance from one or more locations
-  **projections** preserve directions from a particular location.
-  **projections** ensure that area and shape distortion is 'not too bad'.

4.5. Map Projections 2

Purpose

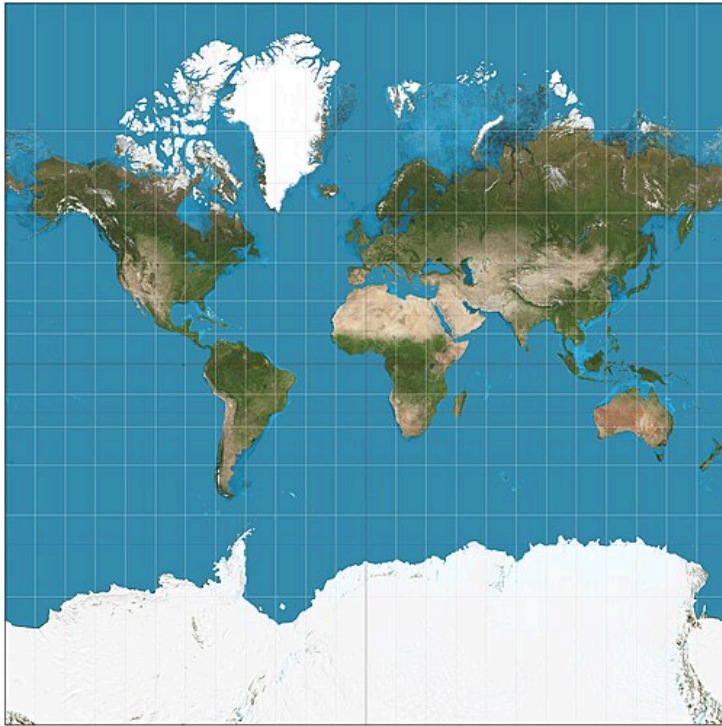
This activity is intended to check your understanding of different map projections.

Task

Complete the following activity which asks you to identify different projections.

Activity: Select the correct answer for each of the projections. Use the   controls to advance to the next projection.

Check your responses using the  **Check** button.



Identify the correct type of map projection

- ☐ Conformal
- ☐ Azimuthal
- ☐ Equidistant
- ☐ Equal area
- ☐ Compromise



4.6. Map fundamentals

Watch the following video which discusses different types of map projections and connects to the next chapter on colour.

Select the advanced player features ^ control to access the video table of contents with chapters targeting each topic.

FIT5147_M3.3 Projections of Data Maps



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Map Fundamentals: Projections and Use of Colour (2022) Monash University <[Panopto Viewer](#)>



Resources

You might be looking for these resources referenced in the video.

Projections

- [Why all world maps are wrong](#) (2016).
- [Map Projection Transitions](#) (Davies n.d.).
- [The true size of](#) (The true size n.d.).

Colour

- [ColorBrewer 2.0](#) (Cynthia Brewer, Mark Harrower and The Pennsylvania State University n.d.)
- [GIS \(Geographic Information System\)](#) (National Geographic 2013).

5. Map Visualisations

Once you have your data, the expected scale of your map, the coordinate system and projection then you are ready for your choice of data map. Here we describe the common map types and knowledge needed for each. Understanding different types of maps, just like different types of tabular data visualisations are important for your ability to determine the most appropriate map type for your data and purpose.

This chapter will describe the many choices and why we might choose one data map (or thematic map as cartographers usually call them) over another. Note whilst this chapter is longer than others they are important elements of each type of map to describe.

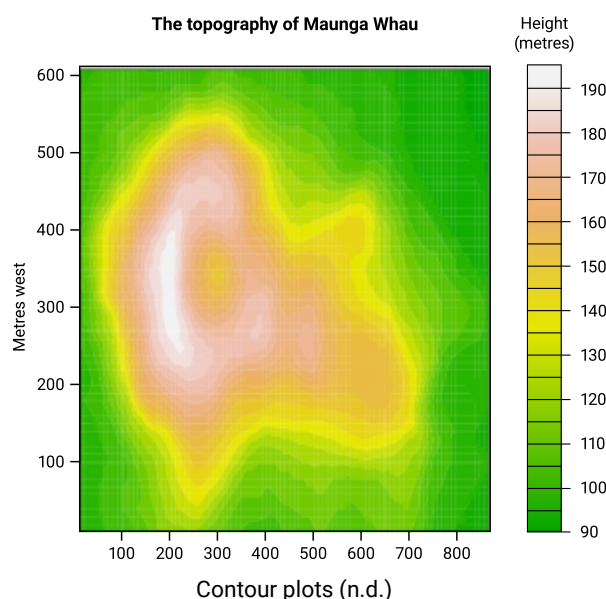
5.1. Common maps and design

The next choice when creating a thematic map is which of the different kinds of visualisation types to use. Select each heading to learn more about the different kinds of maps.

Choropleth and prism	Dot density	Proportional symbol	Contour (isarithmetic) & three-dimensional maps
Cartograms		Flow maps	

Contour (isarithmetic)

A useful metaphor for visualising a scalar field that varies continuously across space is to map it to a kind of conceptual height or elevation for each point on the map. The resulting 3D map is then visualised using standard techniques for showing elevation like contours, wire-frames, surfaces or shaded relief.

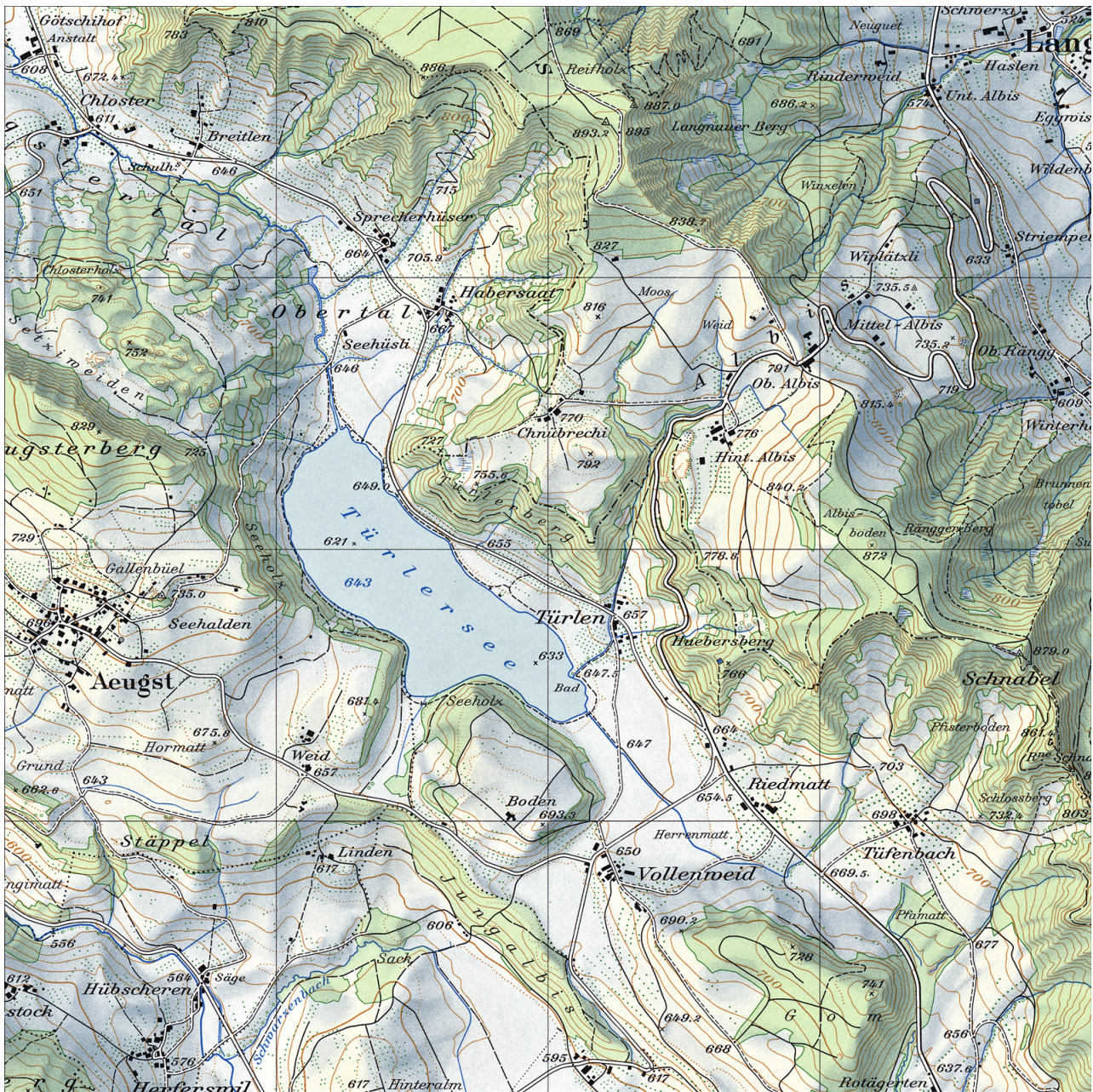


<<https://stat.ethz.ch/R-manual/R-devel/library/graphics/html/filled.contour.html>>

Properly speaking the contours are called *isolines* and the resulting maps are called *isarithmetic* maps. When using contours, it is helpful to add colour tinting to clearly show the regions between each pair of contours.

The main thing to be aware of is that isolines only make sense for data that varies continuously, not data that has abrupt changes or discontinuities. Thus they are ideal for showing temperature and possibly for showing crop yield per acre but not so good for showing the distribution of coal mines.

Contour lines are an essential element to show terrain elevation on topographic maps, i.e. general reference maps. In the example below, contour lines are combined with a shaded relief image that clearly shows the overall shape of the landscape. The contour lines allow for interpolating an elevation value for any terrain location.



Topographic map 1:25,000, Swiss Federal Office of Topography ([swisstopo](https://www.swisstopo.ch)).

5.2. Types of maps

Purpose

This activity is intended to check your understanding of map visualisation types.

Task

Step 1: Complete the following activity which asks you to distinguish between common data maps and designs.

Step 2: For each map, **identify** the visual variables used in the visualisation.

Use the   controls to advance to the next map type. Use the  control to expand to full-screen view.

Check your responses using the  **Check** button.

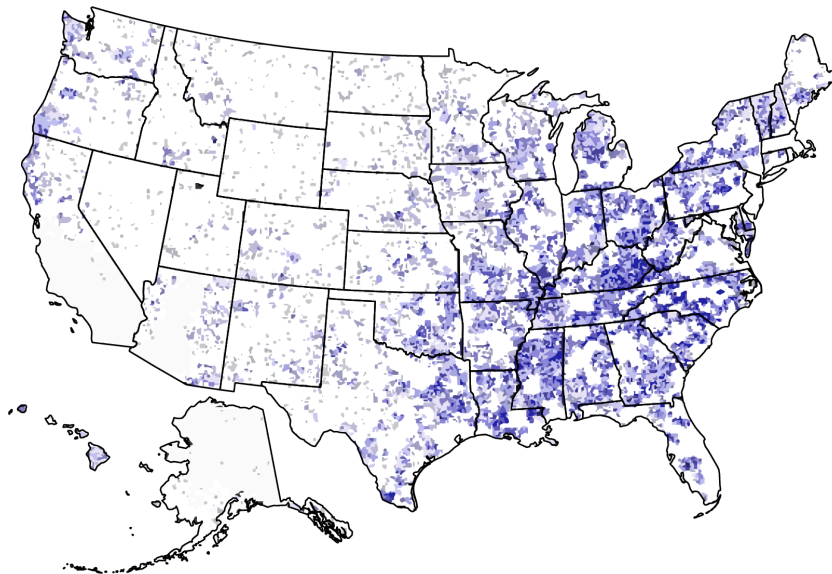


1/8

Map type 1

Where the nonmetro poor are located (2004)

Each dot represents 200 poor persons living in a nonmetro county



Source: computed by ERS using data from the U.S. Census Bureau.

What type of map?

- ☐ Flow map
- ☐ Isarithmic map
- ☐ Cartogram (contiguous)
- ☐ Choropleth map
- ☐ Prism map
- ☐ Proportional symbol map
- ☐ Dot Density map

Identify what visual variables are used (or could be used) in this visualisation

- ☐ Position
- ☐ Shape
- ☐ Size
- ☐ Orientation

☐ Colour - hue☐ Colour - value☐ Colour - saturation☐ Texture/pattern**Check****1 / 8****Map type 1**

5.3. Common Map visualisations

FIT5147 S2 2026 please note: This is a longer video than usual. This is because in S1 and for all prior versions of FIT5147 we dedicated two weeks to spatial data and had more time to discuss map types. The prior page with the data maps per tab and the activity provide the basic information you need to know about map types. If you did not understand those map types or are completely new to you, we recommend this video as providing you extra content. But if you have understood much of the content previously please jump to the next section on the use of colour as this important to understand for all types of map.

Watch the following video which discusses various maps, providing examples of visualisations for each.

Select the advanced player features ^ control to access the video table of contents with chapters targeting each topic.

FIT5147_M3.2 Common Data Maps & Design



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Common data maps and design (2022) Monash University <[Panopto Viewer](#)>



Resources

You might be looking for these resources referenced in the video.

Choropleth maps

- [Choropleth maps](#) (Axis maps 2020).
- [Australian cancer atlas](#) (Australian Cancer Atlas 2023).

Prism maps

- [Prism map vs Choropleth map](#) (Yang, Dwyer, Marriott, Jenny & Goodwin, 2020).

Proportional Symbols

- [Proportional Symbols](#) (Axis maps 2020).

Dot Density

- [Dot Distribution vs Graduated Symbols](#) (GIS Geography 2023).
- [Dot Density Map Example](#) (The New York Times 2010).

Cartogram

- [Covid-19/Coronavirus cases example](#) (Worldmapper 2020).
- [Eleven ways to map a General Election](#) (Stevens 2019).
- [UK general election: live maps and charts](#) (Powell-Smith 2019).
- [Electoral Doctrine: Thirty-nine maps of voting](#) (Worldmapper 2015).

The following resources are good introductions to the design of effective maps.

- [Thematic cartography and geovisualization](#) (Slocum, McMaster, Kessler and Howard 2023).
- [Cartography: Thematic Map Design](#) (Dent, Torguson and Hodler 2009).

5.4. Using colour with spatial data

The number of colours, the sequence of the colour and the type of colour schema are all important for data maps. The wrong type of colour schema for your data can largely mislead the map reader.

What is a colour schema? A colour schema is a collection of colours that is used to encode your data. This is also referred to as colour palette, colour ramp or colour scale.

We tend to use one of three types of colour schema for data maps:

- **Sequential** where usually one colour hue (sometimes multiple) gradually increases from light to dark, e.g., 
- **Diverging** where there is a middle colour and there are two different colour hues on either side of this middle colour, e.g., 
- **Categorical** where there are different colour hues, e.g., 

The choice of type will depend on the data you have.

For instance, is it continuous ordinal data? (choose Sequential)

Or is it numerical, but diverging around an important number that we are looking for our data to be above or below? (choose Diverging)

Or is it categorical, i.e., there is no order to the values? (Choose Categorical)

One of the reasons for only [choosing a few classes](#) for our data is that our eyes cannot perceive too many colours at one time. By default we try not to use more than 5-7 versions of the same hue (i.e., in a sequential schema) and try to limit the number of different hues and shades of hue on the same map. You can also use un-classed maps or continuous colour schemas. See below for future reading on when this is useful.

The difference between each colours in our schema needs to be carefully considered. It can take a long time to choose the right colour schema. Cartographers have done a lot of research for us in this area, e.g., Professor Cynthia Brewer. We, along with many other visualisation specialists and mapmakers, highly recommend using her [colorBrewer](#) schemes, which have been tested for their robustness to be effective on data maps.

Especially be careful of rainbow maps they can mislead the reader, due to a lack of intuitive ordering of the colours and lack of consistency between colour tones and shades. See [eagereyes](#) for more information.

Choose colour vision deficiencies-safe colour schemas where possible. Check the 'colorblind' check box in colorBrewer to remove certain schemas, e.g., avoiding a Red > Green diverging colour schema. This is because there are many people (more prevalent in males) in the world who have [red and green colour deficiencies or deuteranopia](#), where shades of red and green are easily confused.

These can be used with most software packages and are increasingly appearing by default in software. You may need to [create custom colour schemes](#) in Tableau, but there is the standard and well used R [colorBrewer](#) package, Mapbox (see downloads on the [ColorBrewer website](#)), and it can easily be adopted for [d3](#).

Watch the following video which discusses the use of colour in spatial data.

Select the advanced player features ^ control to access the video table of contents with chapters targeting each topic.

FIT5147_M3.4 Using Colour With Spatial Data



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Using colour with spatial data (2022) Monash University <[Panopto Viewer](#)> **Resources**

You might be looking for these resources referenced in the video.

Colour

- [ColorBrewer 2.0](#) (Cynthia Brewer, Mark Harrower and The Pennsylvania State University n.d.).

Rainbow Colour Map

- [How the rainbow color map misleads](#) (Kosara 2013).

Further reading:

- [plotting colour with R](#) - Chapter 10 of "Exploratory Data Analysis with R" (Peng 2020)
- [colour choice for data visualisation](#) - "Which color scale to use when visualizing data" (Muth 2021a)
 - [qualitative or quantitative](#) - "When to use quantitative and when to use qualitative color scales" (Muth 2021b)
 - [diverging v sequential](#) - "When to use sequential and when to use diverging color scales" (Muth 2021c)
 - [unclassified v classed](#) - "When to use classed and when to use unclassified color scales" (Muth 2021d)

6. Analytics for spatial data

Watch the following video which discusses aspects of visualisation and analytics in spatial data.

Select the advanced player features ^ control to access the video table of contents with chapters targeting each topic.

FIT5147_M3.5 Visualisation and Analytics for Spatial Data



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Visualisation and analytics for spatial data (2022) Monash University <[Panopto Viewer](#)>



Resources

You might be looking for these resources referenced in the video.

Choice of Classes

- [The basics of data classification](#) (Axis Maps 2020).
- [When to use classed and when to use unclassed color scales](#) (Muth 2021).
- [Take care of your choropleth maps](#) (Aisch 2011).

Aggregation

- [Modifiable Areal Unit Problem \(MAUP\)](#) (Openshaw 1984).
- [GS-20 - Aggregation of spatial entities and legislative redistricting](#) (Morgan and Evans 2022).

Spatial Autocorrelation

- [Map lineUps: Effects of spatial structure on graphical inference](#) (Beecham et al. 2017).
- [Viel zu warm hier](#) (Blickle et.al. 2019).

Visualising Uncertainties

- [Visualizing uncertain weather](#) (Christiansen 2017).

6.1. Choice of classes

The more classes, the more complex and sometimes confusing a visualisation becomes, but fewer classes can hide or misrepresent the underlying data. When exploring a dataset, using many classes can provide insights, while for presenting results the number of classes should be smaller and designed to show the result you wish to communicate.

A typical number of classes is 4 to 7 though more classes may be used for large dataset. A small number of classes (e.g., 2 or 3 classes) may result in an overly simplified map that hides most of the underlying spatial patterns, whereas a large number of classes (e.g., 8 or more) may make it difficult for the reader to associate a colour on the map with a colour in the legend. Sturges suggests that the number of classes, (**C**), should be:

$$C = 1 + 3.3 \times \log(n)$$

where **n** is the number of observations.

Once you have decided the number of classes, then you have to decide how to partition the observations into these classes: that is the minimum and maximum value for each class. The class intervals (or class breaks) must not overlap and should cover the entire range of the attribute value. This is an example of a one-dimensional clustering problem. Many techniques are used but among the most common are:

- *Equal Interval*: Divide the data range into **C** equally sized intervals.
- *Equal Frequency*: Compute class intervals with the same number of observations. Thus the intervals correspond to quantiles. If there are 4 classes then the classes correspond to the quartiles.
- *Jenks Optimisation*: This chooses the classes to minimise the variance within each class while maximising the variance between the classes. This iterative clustering algorithm is also called *Jenks Natural Breaks* classification.
- *Custom Intervals*: Manually defined class limits can include important values, such as legal thresholds (for example, the maximum legally allowed level of air pollution) or a relevant average value (for example, the national unemployment rate). In addition, "round" and easy-to-remember values can be chosen for custom intervals.

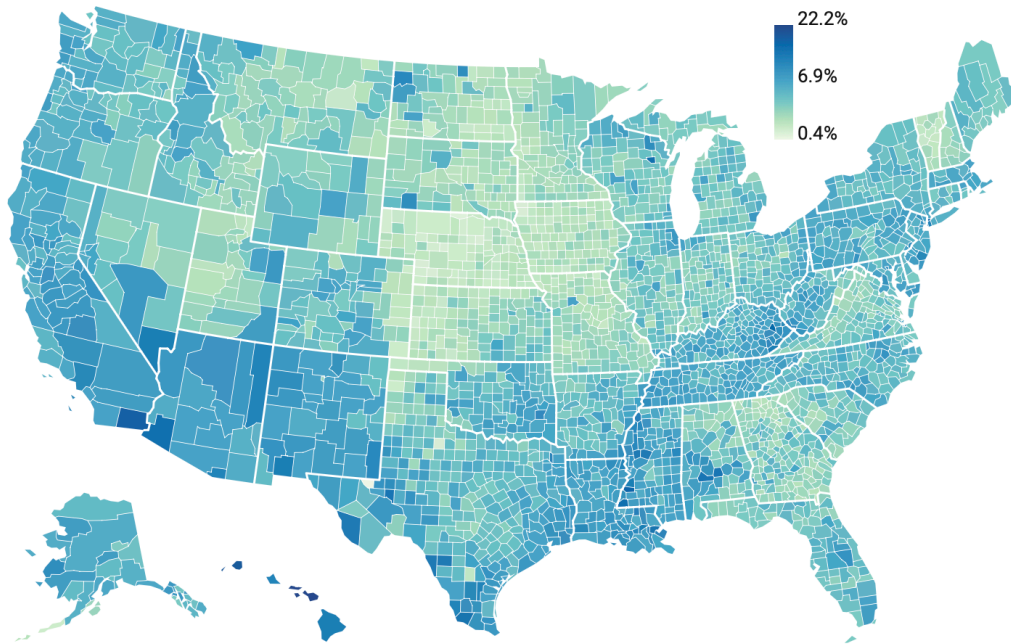
All of these techniques are reasonable approaches. For exploration, Jenks Optimisation probably makes the most sense but it can be difficult to explain to non-data scientists.

Data classification is a type of data transformation. When applied to a quantitative attribute (such as an unemployment rate as on the maps below), classification transforms the quantitative attribute to an ordinal attribute, which are shown in the legend. Of course, when creating an interactive map, you can still show the original quantitative attribute in tooltips and other UI elements.

Data classification is common for choropleth maps, but is also useful for other types of maps, such proportional symbol maps.

Below is a series of maps that illustrates the effect of different classification techniques. All maps show the same attribute (unemployment rate by US counties) and use the same colours. The maps are intended to communicate to a general audience. The first map applies **no classification**, but instead linearly maps the unemployment rate to a linear colour gradient. The map shows nuanced details, but the gradient makes it difficult to reliably associate a colour with a value. The legend indicate the national unemployment rate of 6.9%, but for many counties, it is impossible for the reader to be certain whether the county's unemployment rate is above or below the national average. For these reasons, classification is often useful, as it creates maps that are easier and quicker to read and understand.

Unemployment Rate by US Counties in October 2020

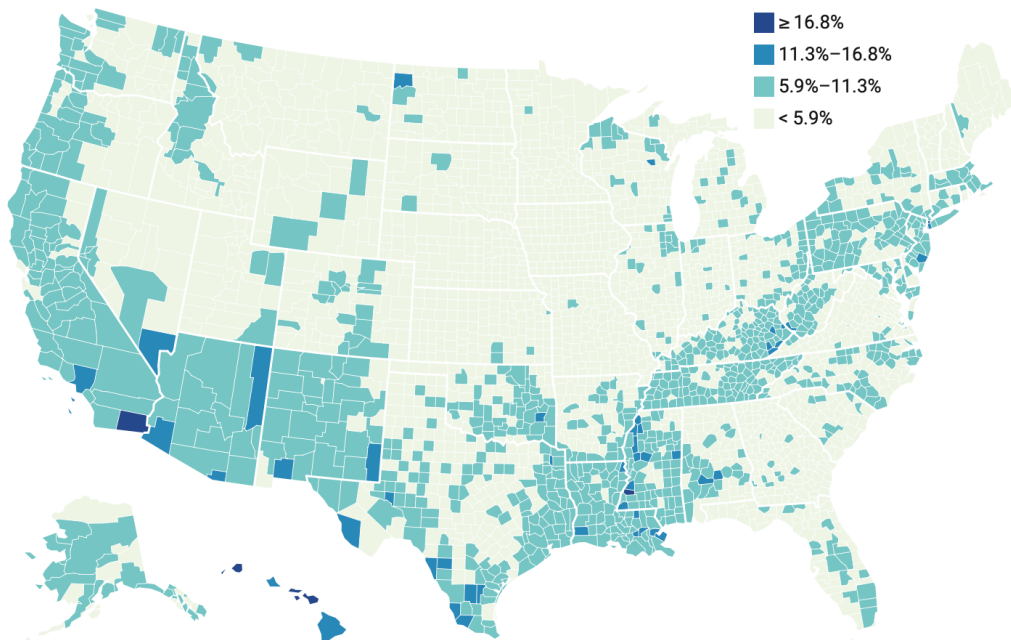


Linear gradient, no data classification

Source: [BLS/LAUS](#) • [Get the data](#) • Created with [Datawrapper](#)

The map below uses four **equal intervals**. There are only a few counties in the two classes with high unemployment rates; the vast majority of counties are in the two classes with low rates. This classification is useful if you need to highlight counties with high rates. However, the class breaks at 5.9%, 11.3% and 16.8% are not intuitive and very difficult to remember. To make this map easier to work with, the rates should be manually set at 5%, 10% and 15%. These class limits would be much easier to remember and would likely result in a visually similar map that communicates the same message.

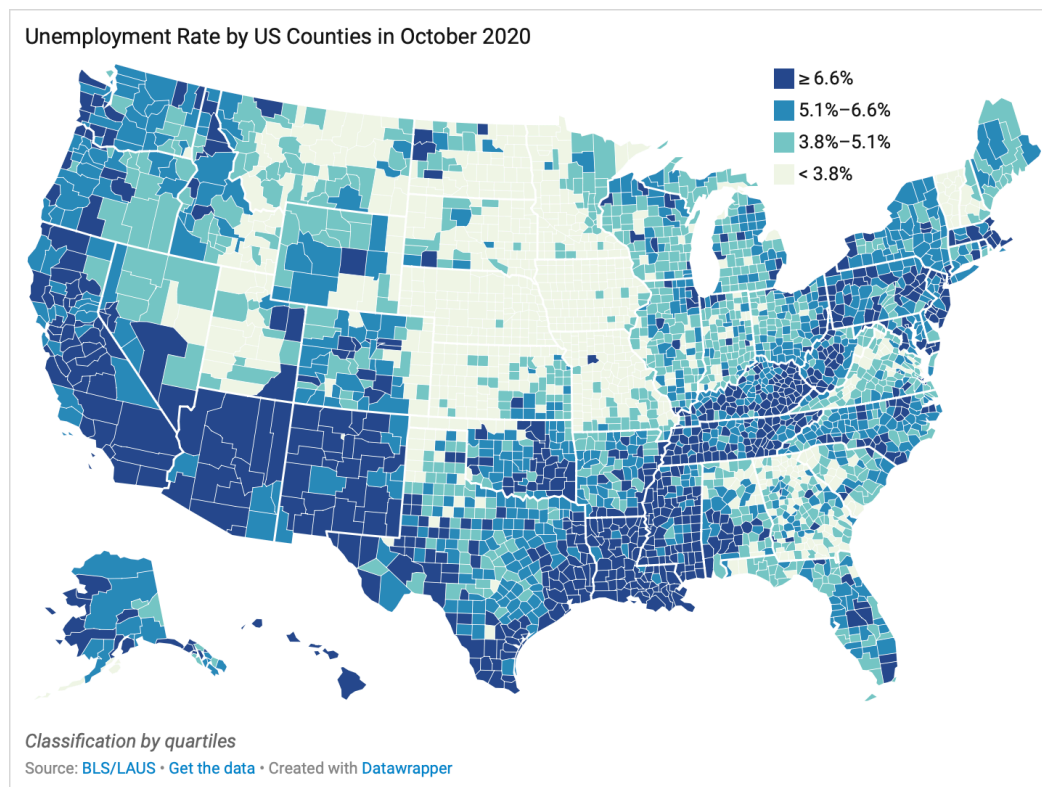
Unemployment Rate by US Counties in October 2020



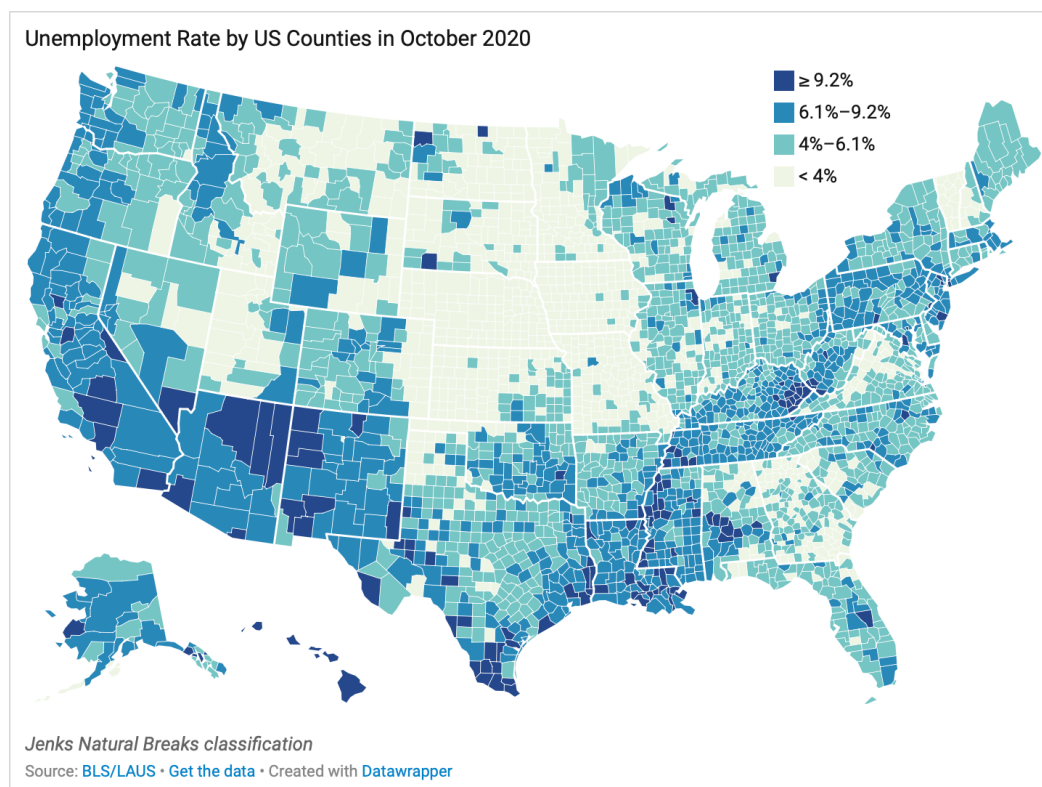
Four classes with equal intervals

Source: [BLS/LAUS](#) • [Get the data](#) • Created with [Datawrapper](#)

The map below classifies unemployment rates by **quantiles** (or quartiles since there are four classes). Each class contains the same number of counties. For the US unemployment rate, the quartile classification results in much more variety across the map, when compared to the equal intervals classification. The quantiles classification enables the reader to compare unemployment rates at a local level and detect interesting local patterns. However, counties with a particularly high employment rate do not stand out on this map, which may or may not be the intention of your map. The quantiles classification also results in difficult-to-remember class breaks (3.8%, 5.1% and 6.6%). Manually changing the breaks to 4%, 5% and 6.5% (or the national average of 6.9%) would result in a map that is easier to read.

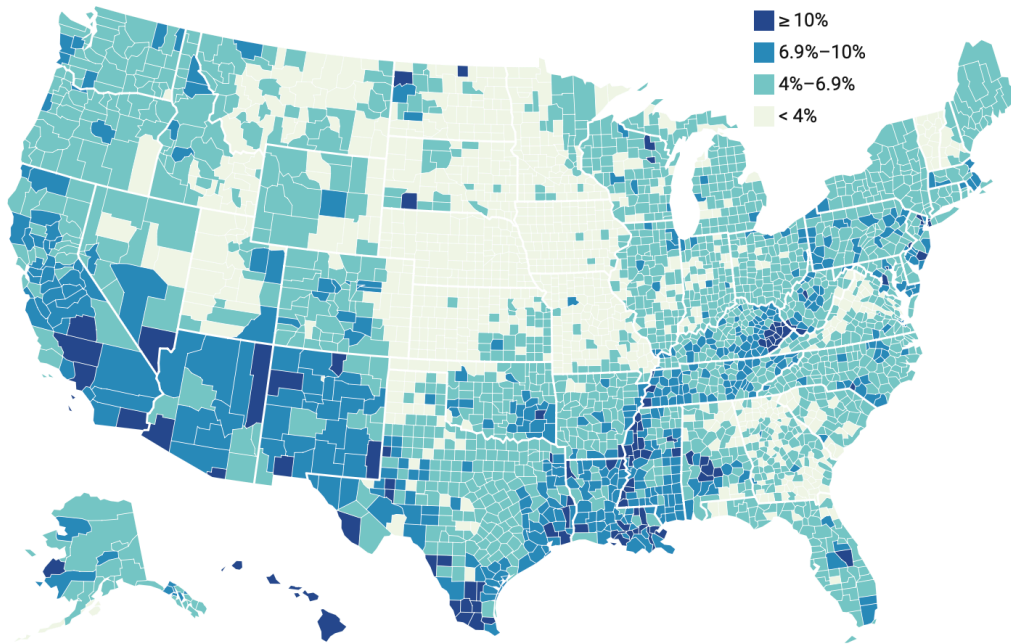


The map below uses the **Jenks Natural Breaks** method with four classes. The Jenks classification minimises the variance within each class while maximising the variance between classes. For this particular dataset, the maps created with quantiles classification and Jenks classification are similar, but the Jenks classification tends to more clearly show regional pockets of high unemployment rate. The comment made about difficult-to-remember class breaks for the two previous maps also applies to this map.



The last map below uses custom manually defined class breaks with four classes. The national average of 6.9% is included, and the two other breaks (4% and 10%) are placed at easy-to-remember values that are both at a distance of approximately 3% from the national average. Again, this may or may not be what you intend to communicate with your map.

Unemployment Rate by US Counties in October 2020



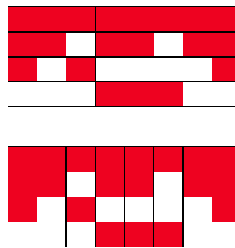
Custom classification with breaks at the national average (6.9%) and easy to remember values of 4% and 10%.

Source: [BLS/LAUS](#) • [Get the data](#) • Created with [Datawrapper](#)

6.2. Aggregation and clustering

A key part of the data map design is to choose the level of detail in which you wish to show the spatial data. Too much detail can overwhelm the viewer while too little detail may mean that the reader is lulled into thinking that the data is more homogeneous than it is. In interactive visualisations, the level of detail can be controlled by the user, either implicitly through zooming or explicitly by a control.

Spatial aggregation—combining data from adjacent regions to form a bigger region can be quite misleading if not done well. This is because the choice of region boundaries can greatly affect the results of aggregation. For example, the boundaries of electoral districts can change the result of elections and gerrymandering is one way politicians can ensure they win an election even though most people do not vote for them.



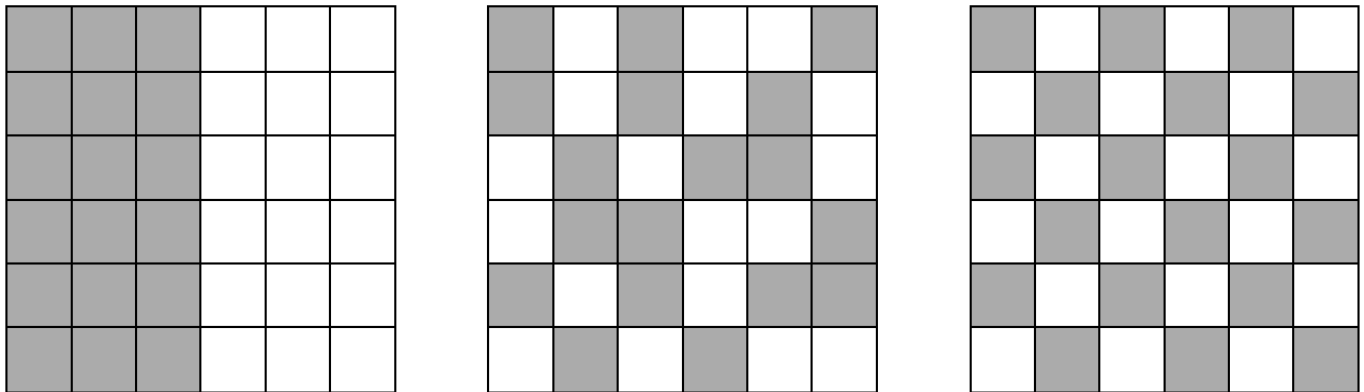
Aggregation (2022) Monash University

If cells are grouped horizontally rather than vertically, aggregation gives a very different result. In this example, horizontally red is the maximum for only 2 groups, but vertically it is the maximum for all groups.

6.3. Spatial autocorrelation

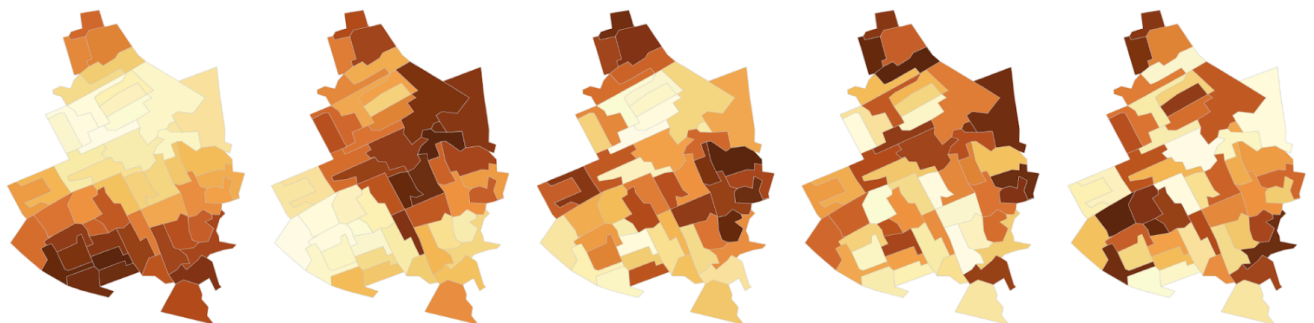
One of the main reasons for visualising data on a map is to see whether data values are correlated with spatial location (called *spatial autocorrelation*). Spatial autocorrelation can also be tested using statistics like the [Moran coefficient \(Moran I\)](#) (Wikipedia 2022) or [Geary's C](#) (Wikipedia 2022). When exploring data, it is wise to confirm the correlation identified visually is statistically meaningful.

The following diagram shows: left: positive spatial autocorrelation; middle: random distribution so no spatial autocorrelation; right: negative spatial autocorrelation.



Spatial autocorrelation (2022) <<https://bit.ly/3QFL60X>>

To gain a better understand of how this conveys on a map. Here is a choropleth map showing what data values across the levels of spatial autocorrelation would look like. The colour legend is the same, but the values of the data differ in this example. Here, the far left map shows high positive spatial autocorrelation; The map shows similar colours clustering together and a clear spatial pattern is revealed on the map. The data appears to smoothly change across the map. Alternatively, the far right shows the highly negative spatial autocorrelation where the data on the map appears to be very different between each region and there is no spatial pattern within the data.



Areas that are next to each other usually have similar colours
So we see more gradual smooth changes across the map
...and the colours look ordered

PROXIMITY
SMOOTHNESS
STRUCTURE

Areas that are next to each other tend NOT to have similar colours
So we see lots of changes across the map
...and the colours look mixed up

Explanation of what spatial autocorrelation looks like on a choropleth map (from Beecham et al, 2017)

References:

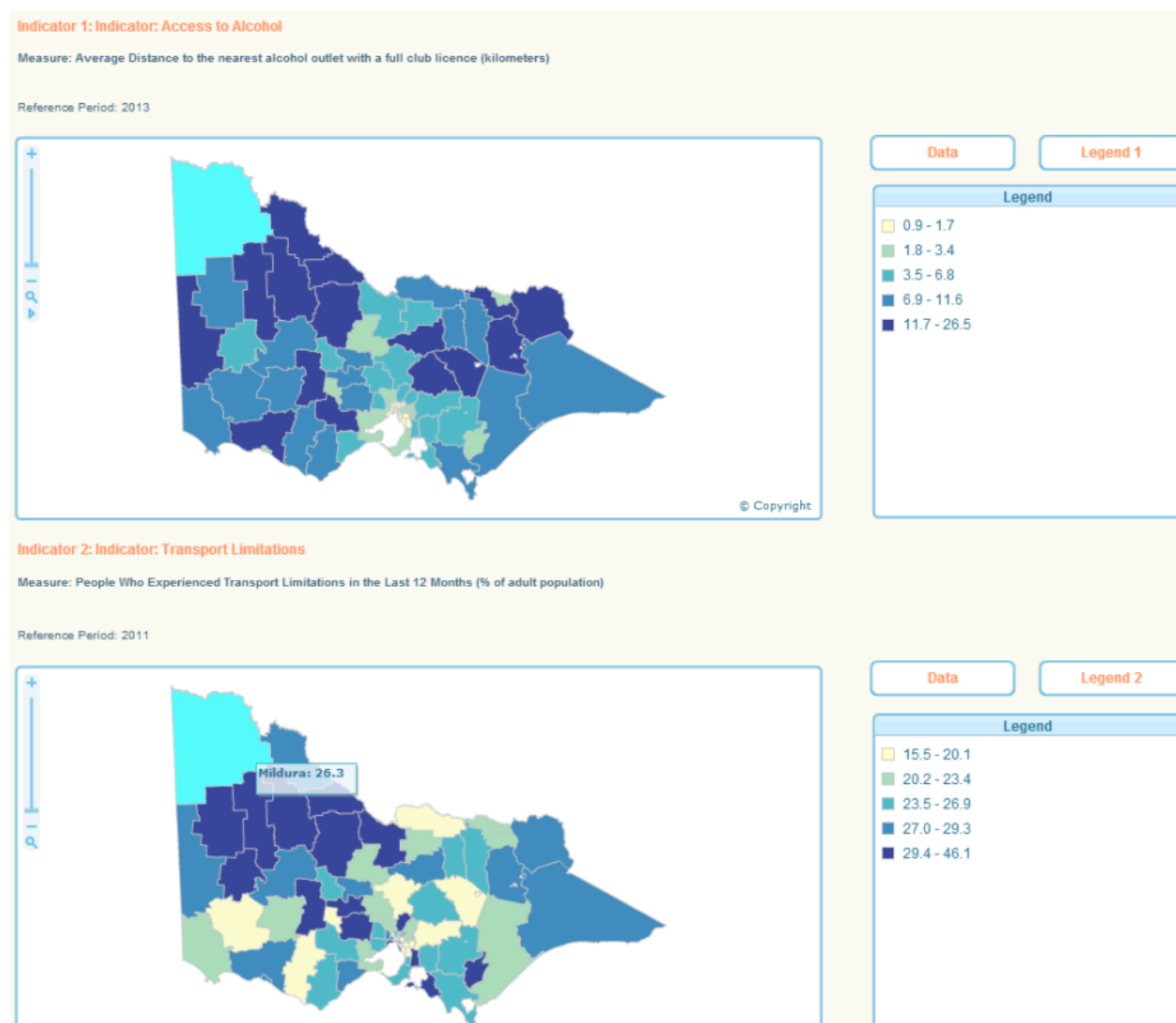
- R. Beecham, J. Dykes, W. Meulemans, A. Slingsby, C. Turkay and J. Wood, (2017) Map LineUps: Effects of spatial structure on graphical inference in *IEEE Transactions on Visualization and Computer Graphics*, vol. 23, no. 1, pp. 391-400, Jan. 2017, doi: [10.1109/TVCG.2016.2598862](https://doi.org/10.1109/TVCG.2016.2598862).

6.4. Multivariate visualisation

We have focused on showing the spatial distribution of a single variable or attribute. However, in most real-world applications data has many attributes and it is the interconnection between these different attributes that the data scientist is most interested in exploring. Unfortunately, this is difficult as showing the data on a data map means that two visual variables ((**x**) position and (**y**) position) are already used up, and other variables must be used to show different attributes.

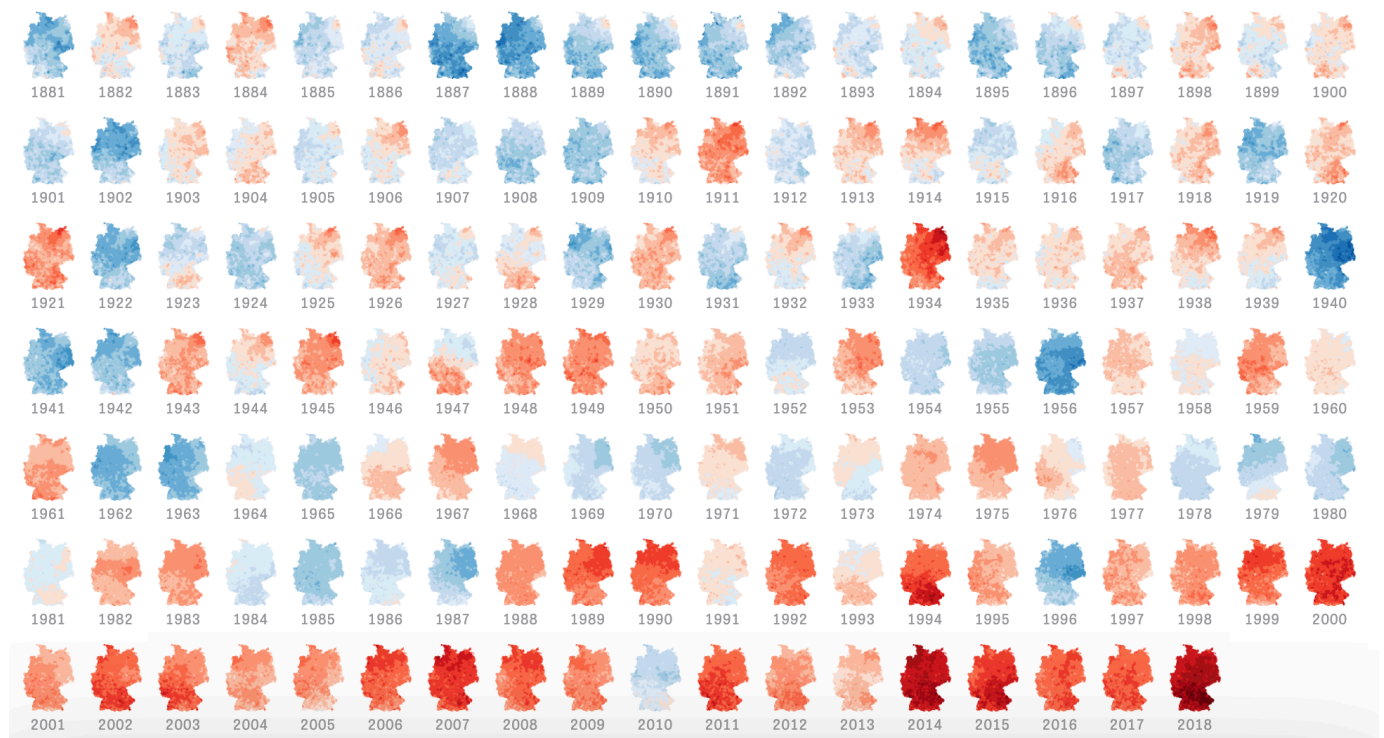
Some approaches are to use two colours or a mix of colour and texture on choropleth maps and more complex glyphs on proportional symbol maps. However, the most common approach is to show multiple maps side-by-side (a technique called 'small multiples') with a map for each attribute, or to overlay two maps. Interaction allows filtering and brushing to show a connection between the different attributes or toggle between the overlays.

The following images show multivariate data with brushing, where Mildura is highlighted in both maps (but not both data point).



Access to Alcohol and Transport (2011) Community indicators consortium

The use of small multiple maps is not just for exploration, but also for data communication and presentation. Here is a good example from Ziet Online on of Germany's popular online newspapers. The story reflects on recent years of spikes of extreme heat and reports on whether what measurement data shows since weather records began in 1881. Heat stripes are shown where blue mean cooler and red means warmer. The stripes expend to maps to present whether these patterns are clear in cities and sections of the map over time.



Small multiple maps to compare if Germany is getting warmer. (Blickle et al, 2019).

It is important for designing maps like these, that the maps use the exact same legend so data is comparable. The different years of data should also be collected in the same way and comparable. Furthermore the maps should be the same size, so patterns seen are comparable.

References:

- Blickle, P., Erdmann, E., Gortana, F., Mast, M., Stahnke, J., and Tröger, J. (2019) Viel zu warm hier (Way too hot here), Climate Change, Zeit Online Newspaper, December 10, 2019, 1:02 p.m. Available at: <https://www.zeit.de/wissen/umwelt/2019-12/klimawandel-globale-erwaermung-warming-stripes-wohnort> Accessed on: 24 Mar 2024.

6.5. Visualising uncertainty

"Being sure is good; being uncertain is not necessarily bad" (Marx, 2013: 613).

One of the disadvantages of visualisations and maps in particular is that they can mask uncertainty and errors in the data and encourage the viewer to have more trust in the patterns than it warrants. Good map design should ensure that the visualisation does not install false confidence in the data or patterns.

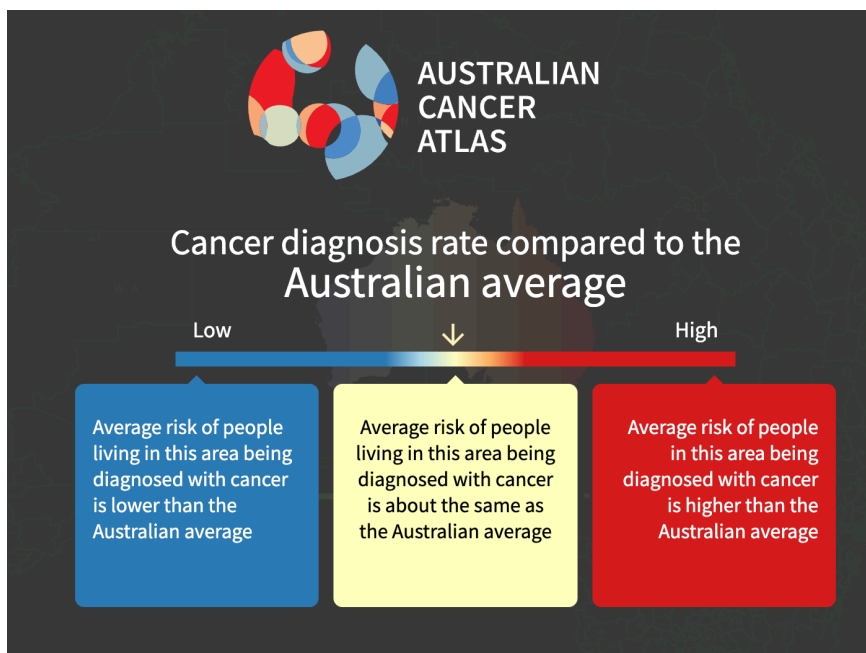
Uncertainty arises in several ways: sampling uncertainty or measurement errors in attribute values and positional accuracy are the most common.

There are two main ways of showing uncertainty. *Intrinsic approaches* tightly couple the visualisation of uncertainty with the attribute being visualised. Specialised visual variables including crispness, resolution and transparency are commonly used in intrinsic approaches. On the other hand, *extrinsic approaches* decouple the visualisation of uncertainty from a visualisation of the attribute and its spatial location. Extrinsic approaches include error bars or a separate map showing confidence levels.

The Australian Cancer Atlas

A good example of both the use of intrinsic and extrinsic approaches in one resource can be found in the [Australian Cancer Atlas](#) (ACA).

The ACA presents the burden of cancer (via two attributes cancer diagnosis and excess deaths) at a fine geographical detail across the whole of Australia. The ACA team specifically created a geostatistical Bayesian model for the purpose. The design of the ACA is based around whether a geographical area has an estimate that is higher or lower than the national average. This is particularly done through the use of a blue-yellow-red diverging colour scheme (yes, from ColorBrewer!):



With all statistical estimates, there is some degree of uncertainty or imprecision. The larger the uncertainty, the less convincing the estimate is of the true value. On the other hand, the smaller the uncertainty, and the further from the Australian average, the more likely it is to have a real difference. Low populated areas in the case of cancer are also often more likely to have more uncertainty.

Here we provide an overview of how uncertainty in the estimates are presented. For more information about model see the ACA documentation (ACA, 2024) and online ebook (Duncan, 2019) resources.

Uncertainty of the model estimates are reported in the ACA using both intrinsic and extrinsic methods. In the case of the ACA, this allows a user (the diverse audience here was aiming to particularly make the data available for researchers, policy makers, practitioners, journalists as well as the general public) to gain an in-depth understanding of the detail if they wish, but also gain an overview of the overall uncertainty across the map. The representation of the model uncertainty was extremely important for the ACA design to ensure the uncertainty of the model was communicated effectively for accurate decision making.

Intrinsic:

The ACA uses a blue-yellow-red diverging colour schema to differentiate between low and high cancer incidence or survival. In order to incorporate uncertainty a yellow transparency is added to the map (this is on by default) based on the posterior probability difference value as it increases from 0 to 1 (see Duncan, 2019 for more information). Using the same colour as the Australian average facilitates a simple interpretation of the estimates. Those estimates that have higher uncertainty are shown as more yellow, so close to the national average, than those with a low uncertainty level which are shown more at their true colour.

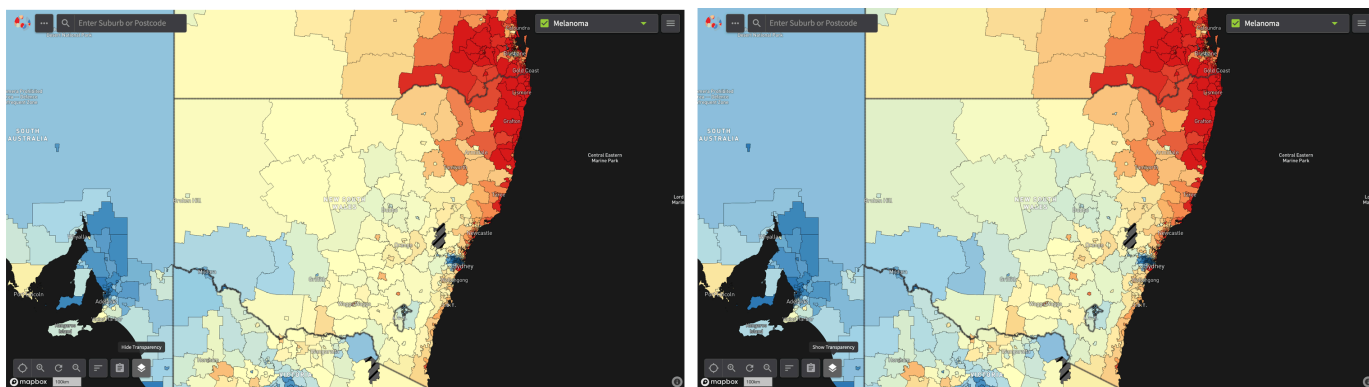


Figure 1: Australian Cancer Atlas with the yellow transparency value ranging from 0-1 based on the model posterior probability difference value (left) switched on and (right) switched off.

Extrinsic:

The ACA has two further visualisations next to the map that were designed specifically to present uncertainty of the individual estimates and allow for detailed comparison of two areas. These plots are referred to by the ACA as the "V plot" and the "wave plot" (hidden by default) and are provided on the right hand side of the ACA map with the main filtering options of the atlas (see the yellow rectangle in Figure 2).

The V-Plot shows whether the estimate represents a significant difference to the average, whilst the wave plot plots the empirical density of the attributes for cancer diagnosis and excess deaths showing the ratio scale density of the model estimates.

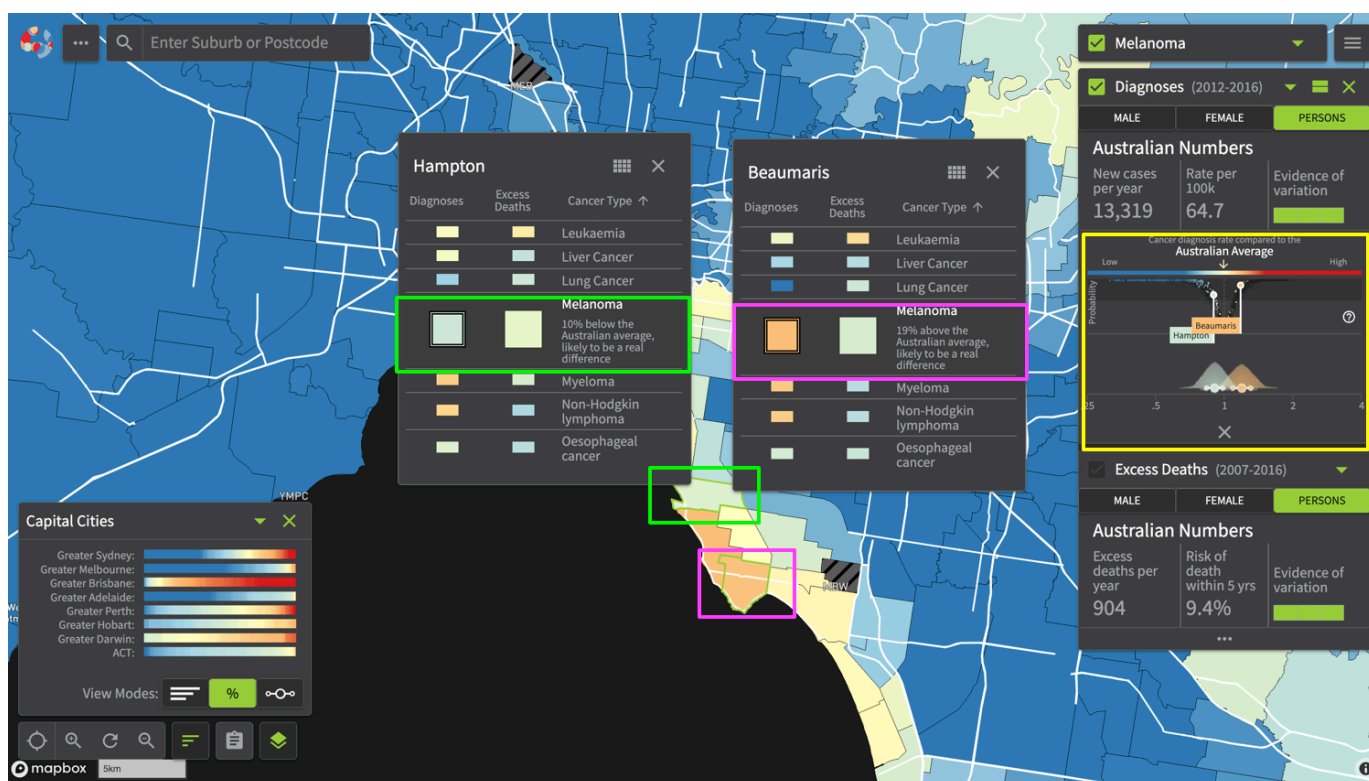


Figure 2: ACA interface showing the map zoomed into Melbourne region. Highlighting two areas on the map in this case Hampton (green small rectangle) and Beaumaris (magenta small rectangle). The colours of the map indicate the value for diagnosis is slightly lower than average for Hampton and slightly higher than average for Beaumaris. The highlighted regions can be seen as highlighted points on the V-Plot and their model estimate distributions can be seen in the wave plot underneath. Colour is consistency used to help connect the information on the screen. A visual comparison of the other cancer models and values is provided in the popup screens for each areas as shown with the larger green and pink rectangles.

Further information about both of these plots can be found explained on the ACA website (ACA, 2024) and online book (Duncan, 2019), particularly Chapter 11 explains the visualisation designs. Notably, these plots and transparency layer were created as part of a long visualisation design process with extensive feedback from the expected audience. The design feedback included removing the axis label of the V plot and wave plot as it caused more confusion to the many users, and the plots alone did enough to portray the difference and inherent uncertainty. Furthermore, all the visuals are interactive and the views are connected. If you interact with the map it updates the v-plot and wave plot. Or if you interact with a point on the v-plot then it highlights on the map. The ACA resource was designed for both visual exploratory analysis as well as communication and presentation tool.

Uncertainty Visualisation is a rapidly changing research field in the data visualisation literature. An [interactive visual survey](#) (Jena, 2020) was recently created to explore some of the main ways in which we are presenting uncertainty in the academic literature. Many of these examples include spatial data.

References:

- ACA (2024), Australian Cancer Atlas, Available at: <https://atlas.cancer.org.au/>, Accessed on 24 Mar 2024.
- Duncan, E. W., S. M. Cramb, P. D. Baade, K. L. Mengersen, T. Saunders, and J. F. Aitken (2019) *Developing a Cancer Atlas using Bayesian Methods: A Practical Guide for Application and Interpretation*. Brisbane: Queensland University of Technology (QUT) and Cancer Council Queensland. Available at: <https://atlas.cancer.org.au/developing-a-cancer-atlas/>, Accessed on 24 Mar 2024.
- Jena, A., Engelke, U., Dwyer, T., Raiamanickam, V. and Paris, C. (2020) Uncertainty Visualisation: An Interactive Visual Survey, 2020 *IEEE Pacific Visualization Symposium (PacificVis)*, Tianjin, China, 2020, pp. 201-205, doi: [10.1109/PacificVis48177.2020.1014](https://doi.org/10.1109/PacificVis48177.2020.1014).
- Marx, V. (2013), *Data Visualization: Ambiguity as a Fellow Traveler*. *Nature Methods*. 10(7): 613-615. doi: <https://doi.org/10.1038/nmeth.2530>

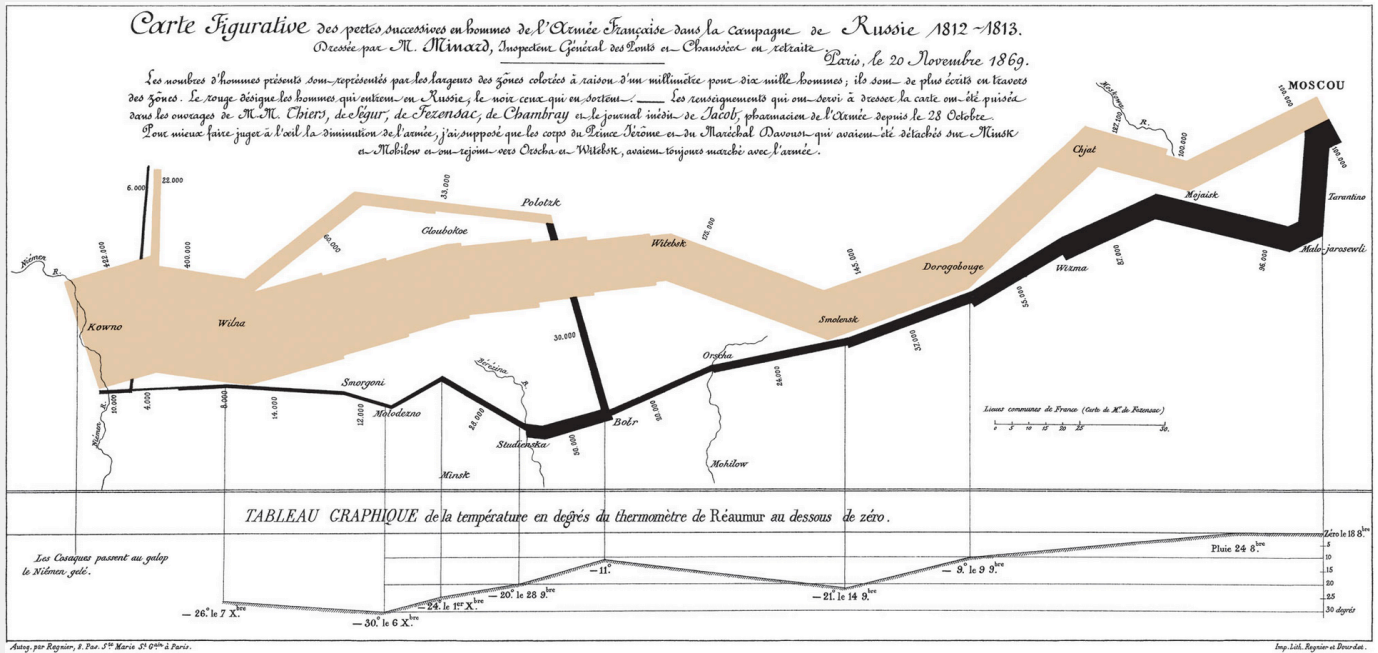
7. End of topic

Well done on reaching the end of the **Topic: Visualising spatial data**.



Reflect

One of the most highly regarded data visualisations of all time is a map by Charles Minard showing Napoleon's march on Moscow during the 1812–1813 Russian campaign.



It uses a flow map style graphic to show the movement of Napoleon's army during the march to Moscow and subsequent retreat. The thickness of the line is proportional to the number of troops. It is a sophisticated multivariate graphic showing the number of soldiers; distance travelled; temperature; troop position and relevant geographic features; the direction of travel; and time taken.

Next

You may now leave this topic book and return to this week's page. Talk to your peers and instructors if you have any thoughts or questions about the concepts, methods and examples mentioned related to this topic.